

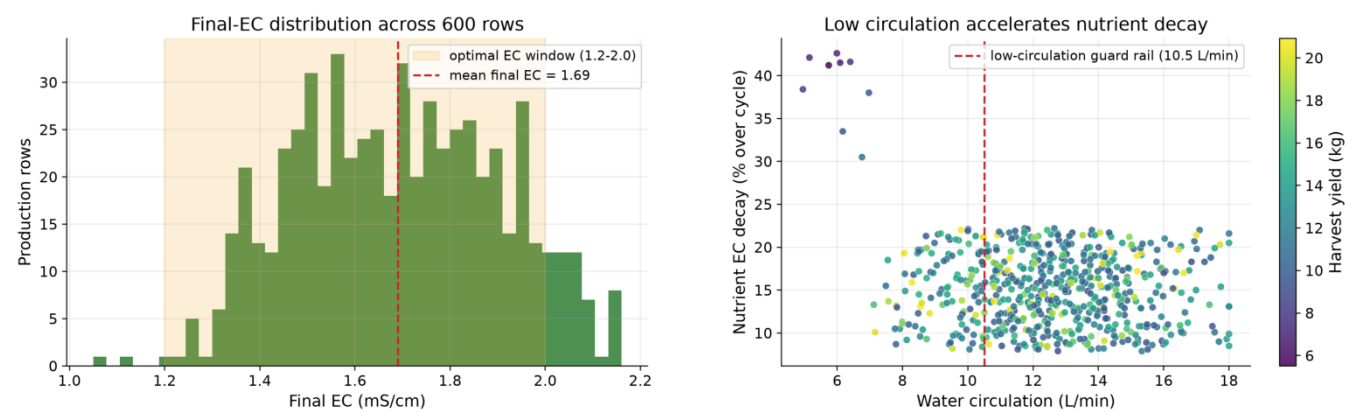
This report analyses 600 production rows of real agricultural automation telemetry and water-chemistry logs across the vertical farm (6 stacks x 10 racks x 10 rows = 60 irrigation channels). The objective is to map where nutrient-solution EC decays, where plant growth stalls, and how circulation, light-cycle hours and humidity drive vertical-stack yields and operating cost.

Key metrics

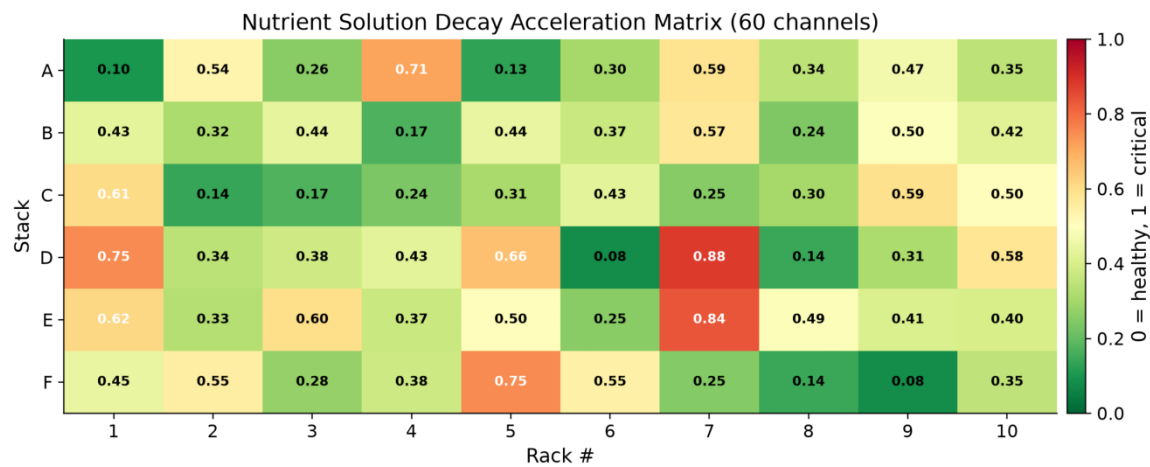
Production rows analysed	600
Irrigation channels (Stack-Rack)	60
Mean final EC	1.69 mS/cm
Rows within optimal EC window	92%
Mean EC decay per cycle	15.3%
Rows with stalled growth	8 (1%)
Low-circulation channels	12 of 60
HIGH-priority decay channels	12
Total harvest yield	7,923 kg

Key findings:

- Water circulation is the dominant lever: channels with low flow show markedly faster EC decay and a higher share of stalled growth.
- Faster nutrient decay drives EC away from the optimal window, which is a leading indicator of stalled growth and lost yield.
- Yield loss is not uniform - it clusters in a small set of channels that can be fixed with targeted maintenance rather than facility-wide change.
- Worst channel is D-R07 (score 0.88, 19% decay, 11.0 L/min).



The decay acceleration matrix aggregates telemetry per irrigation channel (Stack-Rack) and scores each 0-1 by combining nutrient EC decay (45%), circulation deficit (30%), EC-drop magnitude (15%) and growth-stall rate (10%). The grid below shows all 60 channels; hot (red) cells are critical and should be serviced first.



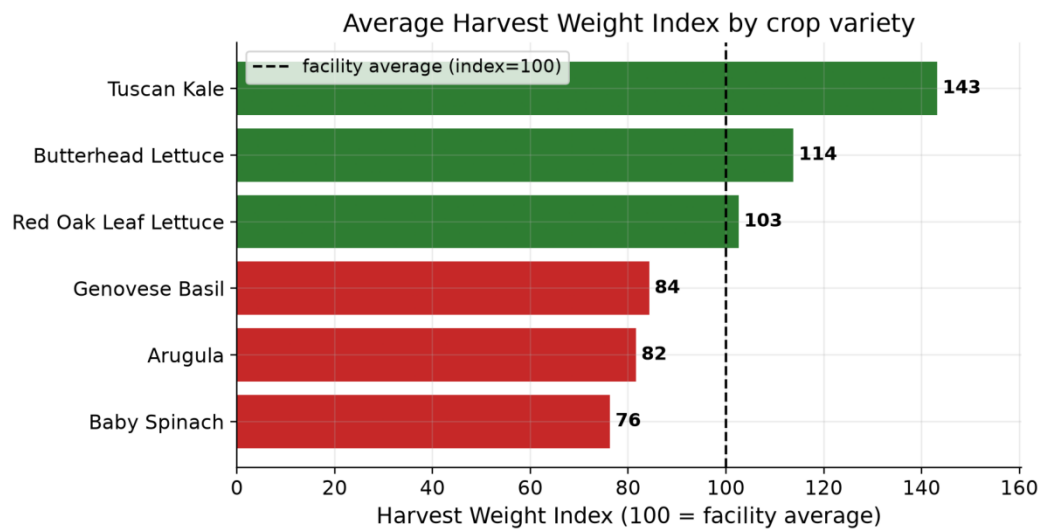
Top decay-acceleration channels (worst 10)

Rank	Channel	Circ L/min	Decay%	Final EC	Harvest kg	Score	Prio
1	D-R07	11.0	18.9	1.66	12.4	0.88	HIGH
2	E-R07	12.6	19.0	1.70	14.1	0.84	HIGH
3	D-R01	12.5	18.2	1.59	14.5	0.75	HIGH
4	F-R05	12.2	17.8	1.72	12.6	0.75	HIGH
5	A-R04	12.3	17.6	1.62	11.7	0.71	HIGH
6	D-R05	12.1	16.9	1.61	11.5	0.66	HIGH
7	E-R01	12.6	16.7	1.66	11.9	0.62	HIGH
8	C-R01	11.4	15.5	1.62	13.6	0.61	HIGH
9	E-R03	11.7	16.9	1.56	15.1	0.60	HIGH
10	C-R09	11.2	16.1	1.71	11.8	0.59	HIGH

Recommended maintenance actions (low-circulation channels)

- 12 channels are flagged for low circulation and need inspection.
- Check for clogged emitters / drippers and biofilm build-up in feed lines.
- Verify circulation-pump duty cycle and impeller wear on flagged channels.
- After restoring flow, flush and re-dose to reset EC into the optimal window.
- Re-measure EC decay per cycle: target a drop back under ~12%.

Worst channel: D-R07 at 11.0 L/min, 19% decay,
0% stalled rows - prioritise this line first.



Average harvest weight index by crop variety

Crop variety	Rows	Avg kg	Median kg	Final EC	Index
Tuscan Kale	95	18.91	18.87	1.70	143.2
Butterhead Lettuce	110	15.02	14.97	1.66	113.8
Red Oak Leaf Lettuce	95	13.54	13.56	1.70	102.6
Genovese Basil	93	11.14	11.10	1.67	84.3
Arugula	94	10.77	10.74	1.72	81.6
Baby Spinach	113	10.08	10.11	1.69	76.3

Optimal nutrient feeding balances (derived from best-yielding rows)

For each crop, targets are the average conditions of its top-quartile (non-stalled) rows.

Crop variety	Target EC	Init EC	Light h	Humid %	Decay%	Exp. kg
Tuscan Kale	1.65	1.99	16.0	66	17.0	20.58
Butterhead Lettuce	1.64	1.93	15.8	65	15.0	16.25
Red Oak Leaf Lettuce	1.71	2.00	15.8	65	14.5	14.78
Genovese Basil	1.72	2.00	16.0	65	14.2	12.19
Arugula	1.75	2.05	16.1	65	14.7	11.73
Baby Spinach	1.71	2.02	16.3	65	15.5	11.05

Feeding-balance principles derived from the data:

- Hold final EC inside the 1.2-2.0 mS/cm window; drift signals decay, not need.
- Dose little-and-often on high-decay channels instead of large batches.
- Fix circulation before increasing nutrient concentration - it is the cheapest lever for both yield and operating cost.
- Match light-cycle hours to the per-crop targets above; excess light raises uptake and accelerates EC drift without proportional yield gains.
- Re-check the worst channels weekly until decay stabilises under target.